Mini project report on

Operating System

Submitted in partial fulfilment of the requirements for the award of degree of

Bachelor of

Technology in

Computer Science & Engineering

UE24CS242B - Operating Systems

Submitted by:

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under the guidance of

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PES University

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DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

FACULTY OF ENGINEERING

PES UNIVERSITY

(Established under Karnataka Act No. 16 of 2013)

Electronic City, Hosur Road, Bengaluru – 560 100, Karnataka, India

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CERTIFICATE

This is to certify that the mini project entitled

Operating Systems

is a bonafide work carried out by

TANISHA DALMIA **PES2UG24CS550**

In partial fulfilment for the completion of forth semester OS Project (UE24CS242B) in the Program of Study Bachelor of Technology in Computer Science and Engineering under rules and regulations of PES University, Bengaluru during the period Jan 2026 – May 2026. It is certified that all corrections / suggestions indicated for internal assessment have been incorporated in the report. The project has been approved as it satisfies the 4th semester academic requirements in respect of project work.

Signature

Prof. Farida Begum

DECLARATION

I hereby declare that the DBMS Project entitled **Operating System UNIT 4 Orange Problem** has been carried out by us under the guidance of **Prof. Farida Begum** and submitted in partial fulfilment of the course requirements for the award of degree of **Bachelor of Technology** in **Computer Science and Engineering** of **PES University, Bengaluru** during the academic semester Jan – May 2026.

TANISHA DALMIA PES2UG24CS550

OUTPUT SCREENSHOTS

1a

```
tanisha-dalmia@tanisha-dalmia-VirtualBox:~/PES2UG24CS550-pes-vcs$ make test_objects
gcc -Wall -Wextra -O2 -c test_objects.c -o test_objects.o
gcc -o test_objects test_objects.o object.o -lcrypto
tanisha-dalmia@tanisha-dalmia-VirtualBox:~/PES2UG24CS550-pes-vcs$ ./test_objects
Stored blob with hash: d58213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
Object stored at: .pes/objects/d5/8213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
PASS: blob storage
PASS: deduplication
PASS: integrity check

All Phase 1 tests passed.
tanisha-dalmia@tanisha-dalmia-VirtualBox:~/PES2UG24CS550-pes-vcs$
```

1b

```
tanisha-dalmia@tanisha-dalmia-VirtualBox:~/PES2UG24CS550-pes-vcs$ find .pes/objects -type f
.pes/objects/2a/594d39232787fba8eb7287418aec99c8fc2ecdaf5aaf2e650eda471e566fcf
.pes/objects/25/ef1fa07ea68a52f800dc80756ee6b7ae34b337afedb9b46a1af8e11ec4f476
.pes/objects/d5/8213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
tanisha-dalmia@tanisha-dalmia-VirtualBox:~/PES2UG24CS550-pes-vcs$
```

2a

```
tanisha-dalmia@tanisha-dalmia-VirtualBox:~/PES2UG24CS550-pes-vcs$ make test_tree
make: 'test_tree' is up to date.
tanisha-dalmia@tanisha-dalmia-VirtualBox:~/PES2UG24CS550-pes-vcs$ ./test_tree
Serialized tree: 139 bytes
PASS: tree serialize/parse roundtrip
PASS: tree deterministic serialization

All Phase 2 tests passed.
tanisha-dalmia@tanisha-dalmia-VirtualBox:~/PES2UG24CS550-pes-vcs$
```

2b

```
tanisha-dalmia@tanisha-dalmia-VirtualBox:~/PES2UG24CS550-pes-vcs$ echo -e "blob 12\0Hello World!" > .pes/objects/d5/8213f5dbe06
29b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
tanisha-dalmia@tanisha-dalmia-VirtualBox:~/PES2UG24CS550-pes-vcs$ xxd .pes/objects/d5/8213f5dbe0629b5c2fa28e5c7d4213ea09227ed02
21bbe9db5e5c4b9aafc12 | head -20
00000000: 626c 6f62 2031 3200 4865 6c6c 6f20 576f  blob 12.Hello Wo
00000010: 726c 6421 0a                             rld!.
tanisha-dalmia@tanisha-dalmia-VirtualBox:~/PES2UG24CS550-pes-vcs$
```

3a

```
tanisha-dalmia@tanisha-dalmia-VirtualBox:~/PES2UG24CS550-pes-vcs$ bash test_sequence.sh
=== PES-VCS Integration Test ===

--- Repository Initialization ---
Initialized empty PES repository in .pes/
PASS: .pes/objects exists
PASS: .pes/refs/heads exists
PASS: .pes/HEAD exists

--- Staging Files ---
Status after add:
Staged changes:
  (nothing to show)

--- First Commit ---
error: commit failed

Log after first commit:
No commits yet.

--- Second Commit ---
error: commit failed

--- Third Commit ---
error: commit failed

--- Full History ---
No commits yet.

--- Reference Chain ---
HEAD:
ref: refs/heads/main
refs/heads/main:
cat: .pes/refs/heads/main: No such file or directory
tanisha-dalmia@tanisha-dalmia-VirtualBox:~/PES2UG24CS550-pes-vcs$
```

3b

```
tanisha-dalmia@tanisha-dalmia-VirtualBox:~/PES2UG24CS550-pes-vcs$ xxd .pes/index | head -20
00000000: 0100 0000 a401 0000 0000 0000 0000 0000  ................
00000010: 0000 0000 0000 0000 0000 0000 0000 0000  ................
00000020: 0000 0000 0000 0000 0000 0000 0000 0000  ................
00000030: 0000 0000 0000 0000 6865 6c6c 6f2e 7478  ........hello.tx
00000040: 7400 0000 0000 0000 0000 0000 0000 0000  t...............
00000050: 0000 0000 0000 0000 0000 0000 0000 0000  ................
00000060: 0000 0000 0000 0000 0000 0000 0000 0000  ................
00000070: 0000 0000 0000 0000 0000 0000 0000 0000  ................
00000080: 0000 0000 0000 0000 0000 0000 0000 0000  ................
00000090: 0000 0000 0000 0000 0000 0000 0000 0000  ................
000000a0: 0000 0000 0000 0000 0000 0000 0000 0000  ................
000000b0: 0000 0000 0000 0000 0000 0000 0000 0000  ................
000000c0: 0000 0000 0000 0000 0000 0000 0000 0000  ................
000000d0: 0000 0000 0000 0000 0000 0000 0000 0000  ................
000000e0: 0000 0000 0000 0000 0000 0000 0000 0000  ................
000000f0: 0000 0000 0000 0000 0000 0000 0000 0000  ................
00000100: 0000 0000 0000 0000 0000 0000 0000 0000  ................
00000110: 0000 0000 0000 0000 0000 0000 0000 0000  ................
00000120: 0000 0000 0000 0000 0000 0000 0000 0000  ................
00000130: 0000 0000 0000 0000 0000 0000 0000 0000  ................
tanisha-dalmia@tanisha-dalmia-VirtualBox:~/PES2UG24CS550-pes-vcs$
```

4a

```
tanisha-dalmia@tanisha-dalmia-VirtualBox:~/PES2UG24CS550-pes-vcs$ ./pes init
echo "test" > file.txt
./pes add file.txt
./pes commit -m "Initial formatted commit"
./pes log
Initialized empty PES repository in .pes/
Committed: 000800000000... Initial formatted commit
commit 4a01a0bab7b03187b4e40c8f485367a500460a5adfbbf4ee0ae3db6b1599b0ac
Author: PES User
Date:   0

    Initial formatted commit


tanisha-dalmia@tanisha-dalmia-VirtualBox:~/PES2UG24CS550-pes-vcs$ ./pes log
commit 4a01a0bab7b03187b4e40c8f485367a500460a5adfbbf4ee0ae3db6b1599b0ac
Author: PES User
Date:   0

    Initial formatted commit


tanisha-dalmia@tanisha-dalmia-VirtualBox:~/PES2UG24CS550-pes-vcs$
```

4b

```
tanisha-dalmia@tanisha-dalmia-VirtualBox:~/PES2UG24CS550-pes-vcs$ find .pes -type f | sort
.pes/HEAD
.pes/index
.pes/objects/4a/01a0bab7b03187b4e40c8f485367a500460a5adfbbf4ee0ae3db6b1599b0ac
.pes/objects/80/ba92e2f423caa58d4a82f0debe469b7424b8e0f7d18464b24c2e4313093e36
.pes/refs/heads/main
tanisha-dalmia@tanisha-dalmia-VirtualBox:~/PES2UG24CS550-pes-vcs$
```

4c

```
tanisha-dalmia@tanisha-dalmia-VirtualBox:~/PES2UG24CS550-pes-vcs$ cat .pes/refs/heads/main
cat .pes/HEAD
4a01a0bab7b03187b4e40c8f485367a500460a5adfbbf4ee0ae3db6b1599b0ac
ref: refs/heads/main
tanisha-dalmia@tanisha-dalmia-VirtualBox:~/PES2UG24CS550-pes-vcs$
```

Final

```
tanisha-dalmia@tanisha-dalmia-VirtualBox:~/PES2UG24CS550-pes-vcs$ make test-integration
gcc -Wall -Wextra -O2 -c index.c -o index.o
index.c: In function ‘index_load’:
index.c:17:5: warning: ignoring return value of ‘fread’ declared with attribute ‘warn_unused_result’ [-Wunused-result]
   17 |     fread(&index->count, sizeof(int), 1, f);
      |     ^~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~
index.c:18:5: warning: ignoring return value of ‘fread’ declared with attribute ‘warn_unused_result’ [-Wunused-result]
   18 |     fread(index->entries, sizeof(IndexEntry), index->count, f);
      |     ^~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~
gcc -Wall -Wextra -O2 -c commit.c -o commit.o
commit.c: In function ‘commit_create’:
commit.c:196:50: warning: unused parameter ‘commit_id_out’ [-Wunused-parameter]
  196 | int commit_create(const char *message, ObjectID *commit_id_out) {
      |                                          ~~~~~~~~~~^~~~~~~~~~~~~
gcc -o pes object.o tree.o index.o commit.o pes.o -lcrypto
=== Running integration tests ===
bash test_sequence.sh
=== PES-VCS Integration Test ===

--- Repository Initialization ---
Initialized empty PES repository in .pes/
PASS: .pes/objects exists
PASS: .pes/refs/heads exists
PASS: .pes/HEAD exists

--- Staging Files ---
Status after add:
Staged changes:
  staged:   file.txt
  staged:   hello.txt

--- First Commit ---
Committed: 000800000000... Initial commit

Log after first commit:
commit d7d9b622f45c4ef56224294f62c77ee26f18a0d6a310e4c70fd7f67518ca8c48
Author: PES User
Date:   0

    Initial commit



--- Second Commit ---
Committed: 000800000000... Update file.txt

--- Third Commit ---
Committed: 000800000000... Add farewell

--- Full History ---
commit 57cb1510dc787c4e90e9abe58896af36c244e2a798bbdd98c82dc800b8f0bf3d
Author: PES User
Date:   0

    Add farewell


commit 6cdd12356dbcf4fdd84b790d0864347f59f838bd470b3306bdcbec39752a669f
Author: PES User
Date:   0

    Update file.txt


commit d7d9b622f45c4ef56224294f62c77ee26f18a0d6a310e4c70fd7f67518ca8c48
Author: PES User
Date:   0

    Initial commit
8◆W


--- Reference Chain ---
HEAD:
ref: refs/heads/main
refs/heads/main:
57cb1510dc787c4e90e9abe58896af36c244e2a798bbdd98c82dc800b8f0bf3d

--- Object Store ---
Objects created:
5
.pes/objects/47/aa7031d895b3d3af8dd4d2421817342d94ed9b59228c81d36e3a36776b3db8
.pes/objects/57/cb1510dc787c4e90e9abe58896af36c244e2a798bbdd98c82dc800b8f0bf3d
.pes/objects/5f/53156830f50d098cf9a69eb392f8e62342cd64ff68fd0728f337ec2b8a38e0
.pes/objects/6c/dd12356dbcf4fdd84b790d0864347f59f838bd470b3306bdcbec39752a669f
.pes/objects/d7/d9b622f45c4ef56224294f62c77ee26f18a0d6a310e4c70fd7f67518ca8c48

=== All integration tests completed ===
tanisha-dalmia@tanisha-dalmia-VirtualBox:~/PES2UG24CS550-pes-vcs$
```

PHASE 5

Q5.1: Implementation of pes checkout <branch>

- Files to change: The .pes/HEAD file must be updated to point to the new branch file (e.g., ref: refs/heads/new-branch).
- Working Directory: The current files in the working directory must be deleted or overwritten to match the file structure (the "Tree") of the target commit.
- Complexity: The complexity arises from safety. You have to ensure that you don't overwrite a user's unsaved/uncommitted changes during the swap. You also have to handle nested directories correctly when reconstructing the file system from the Tree object.

Q5.2: Detecting a "Dirty" Working Directory

- To detect this, you compare the SHA-1 hash of the file currently in the working directory against the hash stored in the index. If the hashes differ, the file is "dirty."
- Furthermore, you compare the index entry to the tree entry of the target branch. If the working directory version is modified but doesn't match the destination branch's version, the checkout must be refused to prevent data loss.

Q5.3: Detached HEAD State

- What happens: Commits are still created, and HEAD updates to point to the new commit hash. However, because no branch (like main) points to these commits, they are "orphaned" once you switch away to another branch.
- Recovery: A user can recover these commits by finding the hash (using git reflog or by looking at the terminal output from when the commit was made) and creating a new branch at that specific hash: git branch <new-name> <commit-hash>.

PHASE 6

Q6.1: Mark-and-Sweep Algorithm

- Algorithm: Use a Mark-and-Sweep approach. Start at all "known" references (all files in refs/heads/). Recursively follow the pointers: Commit -> Tree -> Blobs. Mark every object visited. Finally, delete any object in the object store that was not marked.
- Data Structure: A Bloom Filter or a simple Hash Set would be most efficient for tracking reachable hashes.
- Estimation: You would need to visit at least 100,050 objects (100,000 commits + 50 branch heads) plus all the unique Trees and Blobs they point to.

Q6.2: GC Race Conditions

- The Danger: If GC runs while a commit is happening, GC might see a new "blob" that isn't yet pointed to by a commit and delete it. The new commit would then point to a "missing" object.
- Race Condition: 1. Commit process writes Blob A. 2. GC starts, sees Blob A has no parent, and deletes it. 3. Commit process tries to write the Tree pointing to Blob A. The repo is now corrupted.
- Git's Solution: Git uses a grace period (usually 2 weeks). It won't delete any "unreachable" object unless it is older than a certain timestamp, ensuring that in-progress operations aren't interrupted.